The right-strip comparison from five conditional counts

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We use the five-count reserve of `loss-direct.tex` to bound the full signed Berry–Esseen excess. Interpolating two binomial CDF values is less expensive than comparing to one prescribed integer anchor. In particular the normal error is a signed interpolation remainder of order $1/n$. No geometric budget or displacement bound of order $1/n$ is needed.

Theorem 1. *Let $n \ge 200$ be an integer and $p \in [2/5, 53/100]$, with $q = 1 - p$ and $v = pq$. Let $Y = B + U$, where B is Bernoulli of parameter p, $\mathbb{E}(U \mid B) = 0$, and the conditional noise laws satisfy*

$$-3/5 \le U_0 \le a_0, \qquad -a_1 \le U_1 \le 3/5, \qquad a_0, a_1 \ge 0, \qquad a_0 + a_1 \le 9/125.$$

Put $s = \mathbb{E}U^2$, $\lambda = ns$, $X = (Y - p)/\sqrt{v + s}$ and $\rho = \mathbb{E}|X|^3$. If $3/40 \le \lambda \le 9/25$, then for independent copies $X_1, \dots, X_n$ and every $|H| \le 46/25$,

$$\sqrt{n}\{\mathbb{P}(X_1 + \cdots + X_n \le H) - \Phi(H/\sqrt{n})\} - C_E\rho < -\frac{\lambda}{100}, \qquad C_E = \frac{3 + \sqrt{10}}{6\sqrt{2\pi}}.$$

This is a theorem about every law satisfying the displayed support, centering, variance and threshold hypotheses. Acquiring these hypotheses for an extremal witness is a separate task. The statement concerns the positive signed discrepancy; reflection requires the corresponding hypotheses for the reflected law. It is not an unconditional cutoff theorem for the full iid conjecture.

1 The crossing formula and five binomial masses

Write

$$t = np + H\sqrt{v + s}, \quad k = \lfloor t \rfloor, \quad u = t - k \in [0, 1), \quad w = t - np,$$

and let π_j and F_B be the mass function and CDF of $\mathrm{Bin}(n, p)$. The support hypotheses imply the interior crossing conditions $na_0 \le t < n(1 - a_1)$. Indeed $v \ge 6/25$, $\sqrt{v + s} < 251/500$, and

$$|w| < 231/250 < 47/50. \tag{1}$$

The retained v13 crossing identity gives

$$\mathbb{P}\Big(\sum X_i \le H\Big) = (1 - u)F_B(k) + uF_B(k + 1) + \mathcal{C}(t) - \sum_{i=1}^{n} \pi_i g_i(t), \tag{2}$$

where $g_i \ge 0$ are the kernels in `loss-direct.tex` and

$$\mathcal{C}(t) = \sum_{j=1}^{n-1} (\pi_{j+1} - \pi_j)\left\{\mathbb{E}(t - j - W_j)_+ - (t - j)_+\right\}.$$

Here W_j is the sum of j independent U_1 variables and $n-j$ independent U_0 variables. All lower threshold events are closed.

For $k-1 \le i \le k+3$, (1) gives $|i-np| < 4$. For $z = i/n$, every point on the segment joining z to p lies in $[.38, .55]$, where $z(1-z) > 23/100$. In particular the counts are interior. The explicit finite Stirling remainder

$$\log(m!) = (m+\tfrac{1}{2})\log m - m + \tfrac{1}{2}\log(2\pi) + r_m, \qquad 0 < r_m < 1/(12m),$$

and $D''(z\|p) = 1/(z(1-z))$ give

$$\sqrt{n}\pi_i \ge \frac{\phi(0)}{\sqrt{z(1-z)}} \exp\left\{-\frac{8+1/12}{200(23/100)}\right\} > \frac{199}{250}\left(1 - \frac{8+1/12}{46}\right) > \frac{13}{20}. \tag{3}$$

The same finite Stirling proof as in the retained count appendix is used; no local central limit theorem is invoked. The elementary bounds $199/500 < \phi(0) < 2/5$ follow from $25/8 < \pi < 22/7$.

Let $\alpha = 39/40$, $b = 203/200$ and $x = 3\alpha\lambda$. Taylor's integral remainder gives $1 - e^{-x} \ge x - x^2/2 + x^3/6 - x^4/24$ for every $x \ge 0$. The five-count theorem, monotonicity in n, and (3) yield

$$\sqrt{n}\sum_i \pi_i g_i \ge W(\lambda) := \frac{13}{50}\left\{\frac{x - x^2/2 + x^3/6 - x^4/24}{2} - \frac{(9/25)b\lambda + 3b^2\lambda^2}{16} - \frac{10(22/7)^2}{200}\lambda\right\}. \tag{4}$$

No omitted count needs a new estimate: all other weighted kernels are nonnegative. The finite-window theorem already controls its omitted bump translates, with their entire fourth-moment tail.

2 Curvature and the signed normal interpolation

Use the summable-curvature lemma in the v14 note `geometry-slack.tex`. Set

$$c = (np - q - t)_+, \qquad T = \frac{100}{47}\lambda, \qquad M = \frac{9}{25}T + 3T^2.$$

The actual conditional variance on the increasing side of the binomial mass is at most $\lambda/q \le T$, and its fourth moment is at most M. For $H \le 0$ we have $c < 23/50$; for $H \ge 0$ it is zero. Also

$$\sqrt{T} < \frac{3}{2}\sqrt{\lambda} \le \frac{3}{2}(\lambda + 1/4), \qquad v^{3/2} > 117/1000.$$

The last square-root inequality is $(\sqrt{\lambda} - 1/2)^2 \ge 0$.

The binomial mode coefficient $\beta = 41/100$ in the curvature lemma is valid throughout this right range. To check it, put $N = n+1$ and $r = m/N$ for a mode $m = \lfloor (N+1)p \rfloor$. Since $|m - Np| \le .6$ and $|1-2p| \le .2$,

$$r(1-r) \ge v(1 - 1/N - 2/N^2) > 39v/40 \qquad (N \ge 43).$$

Finite Stirling, discarding its nonpositive entropy and denominator remainders, gives

$$\max_j \mathbb{P}(\mathrm{Bin}(N,p) = j) < \frac{(2/5)\sqrt{40/39}(516/515)}{\sqrt{Nv}} < \frac{41/100}{\sqrt{Nv}}.$$

Thus, writing

$$S(\lambda, c) = \frac{T}{4} + \frac{3c}{2}(\lambda + 1/4) + \frac{27M}{256}\left(\frac{5}{3} + \frac{5c}{4}\right),$$

we have $\sqrt{n}\mathcal{C}(t) \le (41/100)S(\lambda, c)/(nv^{3/2})$.

Schulz's binomial theorem [1, Theorem1, printed p.1] gives, at both integer anchors and for every $0 < p < 1$,

$$\sqrt{n}\{F_B(j) - \Phi((j-np)/\sqrt{nv})\} < C_E\rho_B, \qquad \rho_B = (1-2v)/\sqrt{v}.$$

Convex interpolation preserves this bound. No sharper middle-p reserve is needed here. For $G(z) = \Phi((z-np)/\sqrt{nv})$,

$$G''(z) \le \frac{\phi(0)(np-z)_+}{(nv)^{3/2}}.$$

The signed linear-interpolation remainder therefore satisfies

$$\sqrt{n}\{(1-u)G(k) + uG(k+1) - G(t)\} \le \frac{\phi(0)}{2nv^{3/2}}u(1-u)(u-w)_+.$$

For $H \le 0$, the variance change to $v+s$ is favorable. Using $u(1-u) \le 1/4$, $u^2(1-u) \le 4/27$ and (1), the total normal cost is at most

$$\frac{2/5}{nv^{3/2}}\left(\frac{231/250}{8} + \frac{2}{27}\right).$$

For $H \ge 0$, the interpolation cost is at most $(2/5)(2/27)/(nv^{3/2})$ and the variance change costs at most $\phi(0)ws/(2v^{3/2})$. The total is at most

$$\frac{2/5}{nv^{3/2}}\left(\frac{2}{27} + \frac{231}{500}\lambda\right).$$

These estimates retain both signs of H, every phase and all threshold atoms.

The cubic correction has the favorable sign. Expanding the actual marginal absolute third moment, and using the inward cubic bounds, gives

$$(v+s)^{3/2}(\rho - \rho_B) \ge s\left\{\frac{87}{250} - \frac{13}{16}s\right\} > 0.$$

Here the two conditional supports have fixed signs relative to p; $\mathbb{E}U_0^3 \le a_0\mathbb{E}U_0^2$, $\mathbb{E}U_1^3 \ge -a_1\mathbb{E}U_1^2$, and $(1+y)^{3/2} \le 1 + 3y/2 + 3y^2/8$ for $y \ge 0$. The right side is positive because $s \le 9/5000$.

3 Four rational endpoint comparisons complete the proof

Define two quadratic error polynomials

$$E_-(\lambda) = \frac{(41/100)S(\lambda, 23/50) + (2/5)(231/2000 + 2/27)}{117/1000},$$
$$E_+(\lambda) = \frac{(41/100)S(\lambda, 0) + (2/5)(2/27 + 231\lambda/500)}{117/1000}.$$

The preceding estimates bound the full signed excess strictly above by $-W(\lambda) + E_\pm(\lambda)/n$, according to the sign of H. It remains to prove positivity of

$$P_\pm(\lambda) = W(\lambda) - E_\pm(\lambda)/200 - \lambda/100.$$

Both are quartics. If c_j denotes their coefficient of λ^j, direct rational expansion gives, in both cases,

$$c_2 < -3/5, \qquad c_3 < 11/20, \qquad c_4 < 0.$$

Consequently on $[3/40, 9/25]$,

$$P''_{\pm}(\lambda) = 2c_2 + 6c_3\lambda + 12c_4\lambda^2 < -\frac{6}{5} + \frac{33}{10}\frac{9}{25} < 0.$$

The minimum of each concave polynomial is attained at an endpoint. The four exact rational comparisons, rounded down only for display, are

	$\lambda = 3/40$	$\lambda = 9/25$
P_-	$> 6075/10^6$	$> 2577/10^6$
P_+	$> 11528/10^6$	$> 11247/10^6$.

The companion checker reconstructs the polynomials and verifies these comparisons using fractions. It performs no subdivision or optimization. Thus $P_{\pm} > 0$ on the entire closed noise interval. Since $E_{\pm} > 0$ and $n \geq 200$, Theorem 1 follows.

Dependencies and remaining range

The new local inputs are the five-count reserve in `loss-direct.tex`, the v13 crossing identity and the v14 summable-curvature lemma. The external probability input is precisely Schulz's all-p binomial theorem; finite Stirling has its explicit retained elementary proof. No geometric-budget, selected-count transport, marked crossing, Poisson comparison or new parameter certificate is used in this larger-noise comparison.

For acquired witnesses, v13 geometry supplies $\lambda < .36$, while v12 handles smaller positive noise. That small-noise theorem may choose a different integer anchor; this causes no problem because the binomial theorem holds at every anchor. Zero noise is Bernoulli. The support strips, raw-threshold bound, sign-specific acquisition and independent finite base remain separate obligations. No statement about unrestricted C_n is made.

References

[1] J. Schulz, *The optimal Berry–Esseen constant in the binomial case*, doctoral dissertation, Universität Trier, 2016. `https://d-nb.info/1197702695/34`.